

Free tools to explore speech AI

Run the unplugged activity first. Then have students test real systems with the same speaker cards and compare word error rates.

STAGE	TOOL	TRY THIS WITH STUDENTS	ACCESS AND PRIVACY
HEAR speech → text	Windows Live Captions vs. Windows voice typing Microsoft	Read the speaker cards aloud to each and score the word error rate. Which copes better with whispering or music? No Windows PC? Phone or Chromebook voice typing works too.	Built into Windows 11. Live Captions (Win + Ctrl + L) generates captions on the device. Voice typing (Win + H) uses online recognition. A live on-device vs. cloud comparison.
SAY text → speech	Read Aloud in Microsoft Edge Microsoft	Have the computer read the speaker cards in different voices. Which words sound robotic, and why?	Free in the Edge browser (Ctrl + Shift + U). No account needed.
MEASURE the S/D/I idea	Reading Coach and Reading Progress Microsoft	Reading Coach: students read aloud, then practice the words the AI detected they struggled with. Reading Progress (for teachers) flags mispronunciations, omissions and insertions: the worksheet's swapped, missing and extra words, in a real product.	Reading Coach: free on the web (coach.microsoft.com) or Windows with a school or personal Microsoft account. Reading Progress: in Microsoft Teams for Education. Both process student voices, so use school accounts and policies.
BUILD train a model	Teachable Machine Google	Train a clap, snap or silence classifier, then test whether it works equally well for everyone's clap.	Free in the browser, no account needed: teachablemachine.withgoogle.com
PLAY AI and fairness	Minecraft Education: Hour of AI Microsoft · Code.org	"The First Night": train an AI agent to survive. "Generation AI": fix a sensor that is unfair about height, a fairness lesson that pairs with "voices for everyone."	Hour of AI lessons are free to play or host; Generation AI is a free demo: aka.ms/hourofai
CREATE teacher prep	Microsoft 365 Copilot Chat Microsoft	Teachers: ask it for new sound-alike sentence cards at your students' level, then check each one yourself.	With Microsoft 365 Education licenses. Students 13–17 only if school IT turns it on; students under 13 are blocked.
GO DEEPER grades 11–12	Building Speech AI companion code open source	Run real speech models (Whisper, wav2vec 2.0) in pre-executed Python notebooks, and read every output before installing anything.	Free (MIT license): github.com/prdeepakbabu/building-speech-ai

Learn more (teachers)

- **Microsoft Learn, "Introduction to AI speech concepts"**: a free short module on speech recognition and synthesis. learn.microsoft.com/training/modules/introduction-ai-speech
- **Microsoft Elevate for Educators**: free AI training, community and credentials for teachers (search "Microsoft Elevate for Educators").
- **IEEE TryEngineering**: free engineering lesson plans. tryengineering.org
- **Code.org Hour of AI**: one-hour AI activities, held around CS Education Week in early December. code.org/en-US/hour-of-ai

Free credits to experiment (teachers and older students)

- **Azure Speech free tier**: 5 hours of speech-to-text and 0.5 million characters of text-to-speech a month (adult Azure account).
- **Azure for Students**: \$100 of Azure credit a year with no credit card, for full-time college students 18 and older.
- **GitHub Education**: free GitHub Copilot for verified students (13 and older) and teachers.

Before students use any online tool: follow your school's technology and privacy policies. Prefer school-managed accounts, and don't upload recordings of students to services your school hasn't approved.